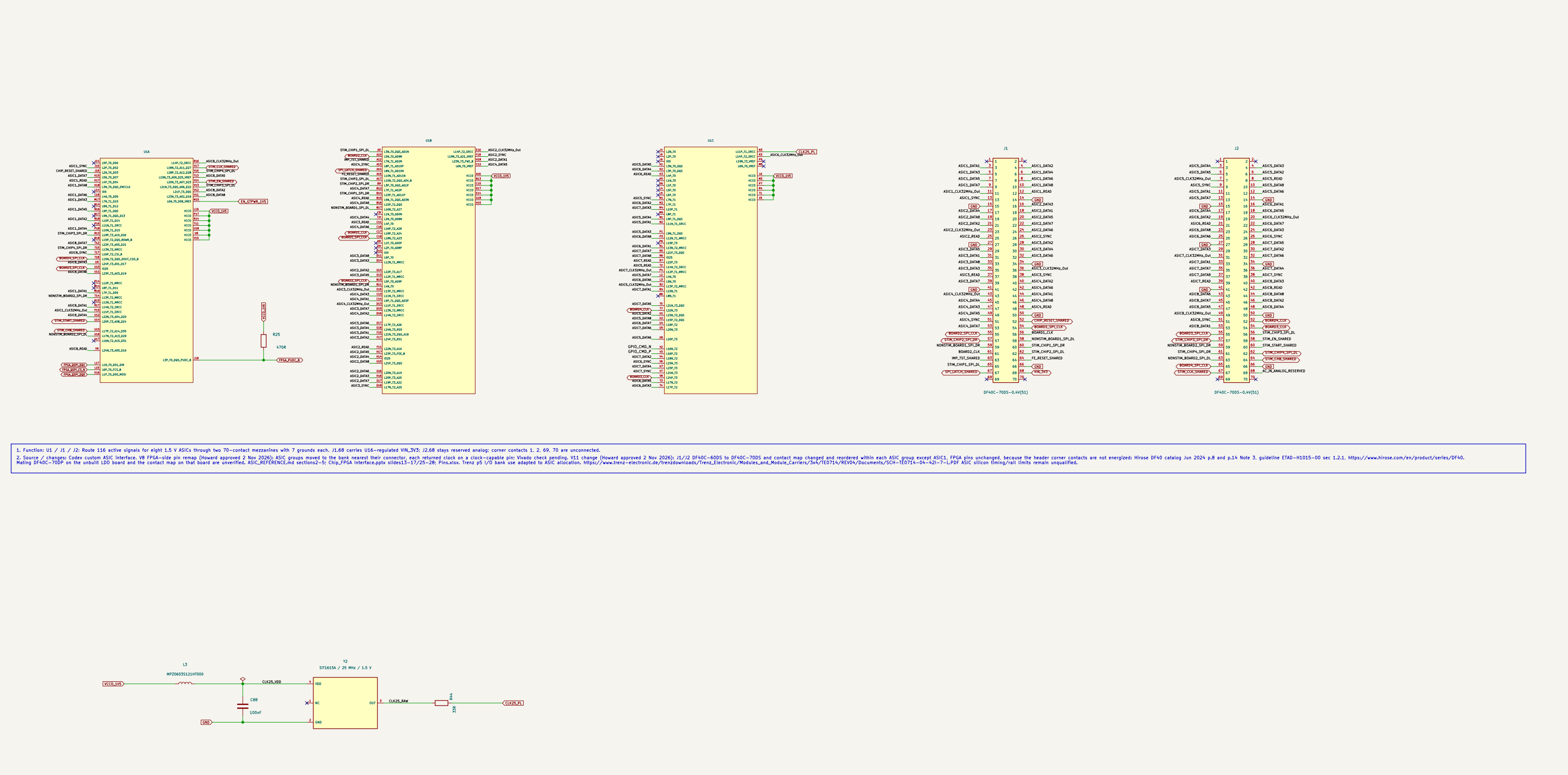
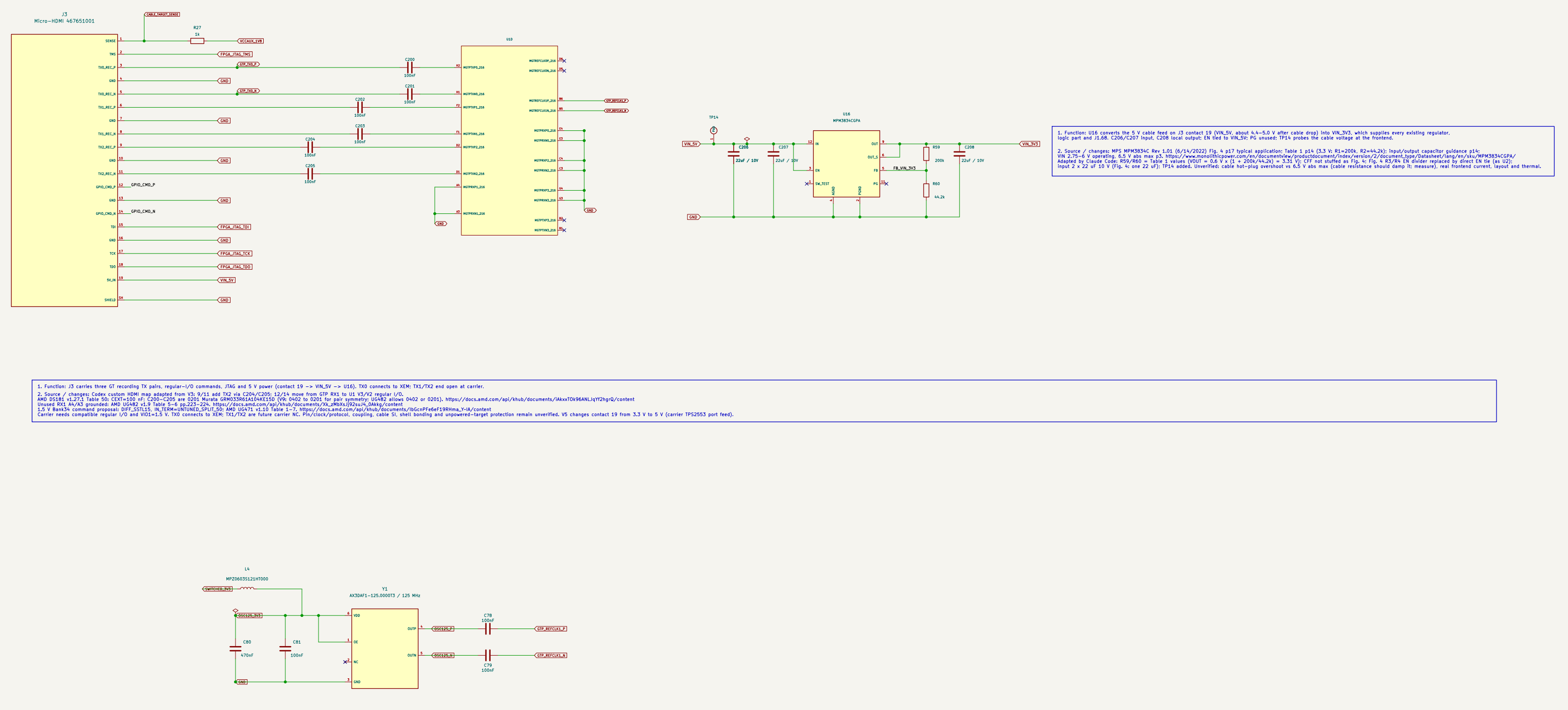
01. FPGA I/O + ASIC INTERFACES + 25 MHz CLOCK | Trenz p5



1. Function: U1 / J1 / J2: Route 116 active signals for eight 1.5 V ASICs through two 70-contact mezzanines with 7 grounds each. J1.68 carries U16-regulated VIN_3V3; J2.68 stays reserved analog; corner contacts 1, 2, 69, 70 are unconnected.
2. Source / changes: Codex custom ASIC interface. V8 FPGA-side pin remap (Howard approved 2 Nov 2026): ASIC groups moved to the bank nearest their connector, each returned clock on a clock-capable pin; Vivado check pending. V11 change (Howard approved 2 Nov 2026): J1/J2 DF40C-60DS to DF40C-70DS and contact map changed and reordered within each ASIC group except ASIC1. FPGA pins unchanged, because the header corner contacts are not energized: Hirose DF40 catalog Jun 2024 p.8 and p.14 Note 3, guideline ETAD-H1015-00 sec 1.2.1. https://www.hirose.com/en/product/series/DF40. Mating DF40C-70DP on the unbuilt LDO board and the contact map on that board are unverified. ASIC_REFERENCE.md sections2-5; Chip_FPGA interface.pptx slides13-17/25-28; Pins.xlsx. Trenz p5 I/O bank use adapted to ASIC allocation. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF ASIC silicon timing/rail limits remain unqualified.

1. Function: Y2:25MHz fabric clock drives N3 SRCC at1.5V. L3 filters1.5V; C88=100nF; R44=33R output damping.
2. Source / changes: Trenz p5 U8/L2/C24/R23 structure adapted: SIT1615 1.5V replaces SIT8008 1.8V; CLK25 uses N3 SRCC. C88 retained100n per SiTime; R44 exact Trenz RC0201FR-0733RL restored. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF https://www.sitime.com/datasheet/SiT1615 Timing/firmware constraints remain unverified.

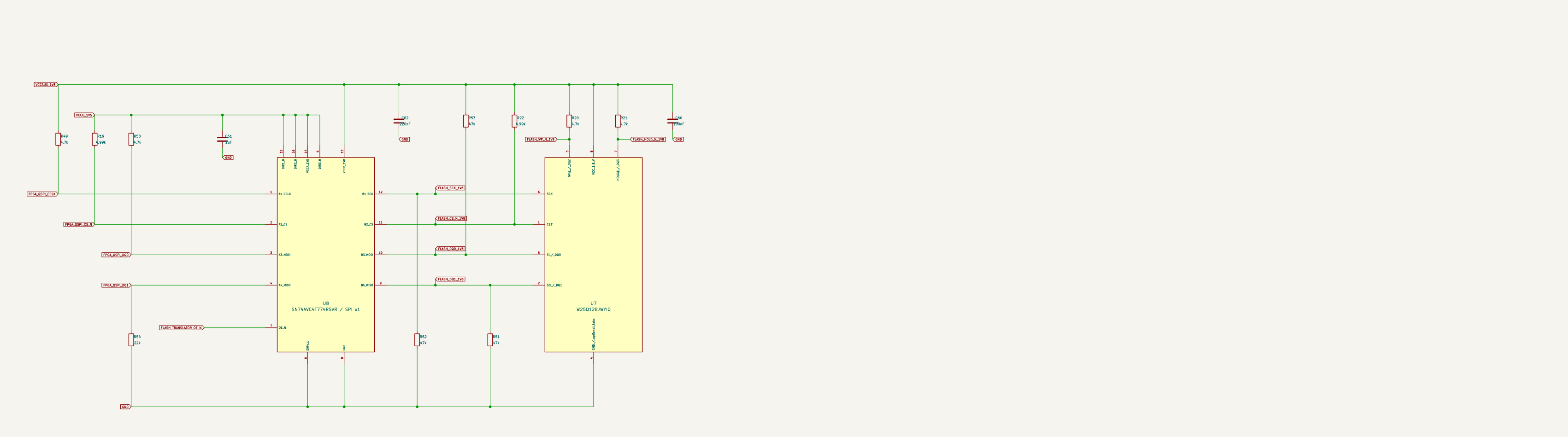
02. GTP CHANNELS + 125 MHz REFERENCE | Trenz p6



1. Function: J3 carries three GT recording TX pairs, regular-I/O commands, JTAG and 5 V power (contact 19 -> VIN_5V -> U16). TX0 connects to XEM; TX1/TX2 end open at carrier.
2. Source / changes: Codex custom HDMI map adapted from V3; 9/11 add TX2 via C204/C205; 12/14 move from GTP RX1 to U1 V3/V2 regular I/O.
AMD DS181 v1.27.1 Table 50: CEXT=100 nF; C200–C205 are 0201 Murata GRM033R61A104KE15D (V9: 0402 to 0201 for pair symmetry; UG482 allows 0402 or 0201). https://docs.amd.com/api/khub/documents/iAkxxTOk96ANLJqYf2hgrQ/content
Unused RX1 A4/A3 grounded: AMD UG482 v1.9 Table 5-6 pp.223-224. https://docs.amd.com/api/khub/documents/Xk_zMbXsJj92suJ4_0Akkg/content
1.5 V Bank34 command proposal: DIFF_SSTL15, IN_TERM=UNTUNED_SPLIT_50; AMD UG471 v1.10 Table 1-7. https://docs.amd.com/api/khub/documents/lbGcnPFe6eF19RHma_Y-lA/content
Carrier needs compatible regular I/O and VIO1=1.5 V. TX0 connects to XEM; TX1/TX2 are future carrier NC. Pin/clock/protocol, coupling, cable SI, shell bonding and unpowered-target protection remain unverified. V5 changes contact 19 from 3.3 V to 5 V (carrier TPS2553 port feed).

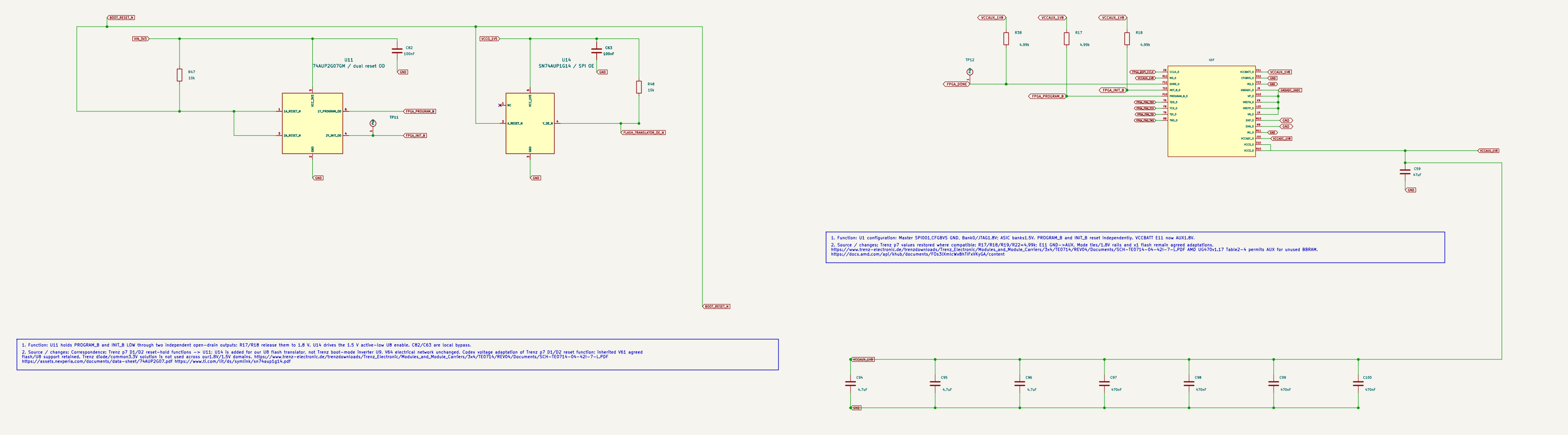
1. Function: Y1 supplies125MHz LVDS through C78/C79=100nF to REFCLK1P/N. L4 filters switched3.3V; C80=470nF and C81=100nF bypass it. The REFCLK input's internal 100R is the only load; OE HIGH, pin2 NC.
2. Source / changes: Trenz TE0714-04-42I-7-L REV04 p6:125MHz LVDS U2, L1 ferrite, C10=470nF, R1=100R and C134/C135=100nF coupling. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF Adapted by Codex: Y1=AX3DAF1-125.0000T3 (exact Trenz oscillator MPN undisclosed). C80=470nF. C81 10uF->100nF; added L4. External R56 removed per UG482 Fig. 5-8 p226 (100nF caps only); with it, AX3 swing is 250–450mV against DS181 350mV minimum. Abracon AX3 rev25Feb2022 pp3,9,12-13 requires active-HIGH OE and shows100nF local bypass. https://abracon.com/datasheets/AX3.pdf REFCLK1 B6/B5 matches Trenz: P-to-P/N-to-N retained. AMD UG482 v1.9 pp225-227,235: LVDS AC-coupled reference. https://docs.amd.com/v/u/en-US/ug482_7Series_GTP_Transceivers Clock jitter, filtered-rail noise, layout and hardware startup remain unqualified. R56 uses the same100R0.1%25ppm candidate as R16; exact source MPN is undisclosed.

03. FLASH + SPI TRANSLATION | Trenz p6 adaptation



1. Function: U7/U8: 16MiB 1.8V SPI NOR flash (Winbond W25Q128JWYIQ, 21-ball WLCSP, 0.5mm pitch) and SN74AVC4T774RSV SPIx1 translation. Bank14 remains1.5V; initialCONFIGRATE3MHz/QE=1 fixed/3-byte addresses.
2. Source / changes: Package changed V26 (Howard 3 Nov 2026): WSON-8 W25Q128JWPIQ -> WLCSP-21 W25Q128JWYIQ, same W25Q128JW die, pins and Q option: QE=1 fixed, so WP#/HOLD# functions are off. W25Q128JW Rev.H 17Apr2023; WLCSP ball map Fig.1e p8, ball table 3.9 p8, WLCSP outline 10.7 p73, ordering pp75-76 (special order). https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/8919/256_W25Q128JW.pdf TI translator: https://www.ti.com/lit/ds/symlink/sn74avc4t774.pdf R19/R22 4.99k; pullups 1.8V. Codex/Claude adaptation of Trenz boot structure: no circuit value changed. Land pattern (0.25mm NSMD), body 3.0x3.0mm and via-in-pad are Claude assumptions; JLC via-in-pad, sou

04. CONFIGURATION + RESET OUTPUTS | Trenz p7 notes

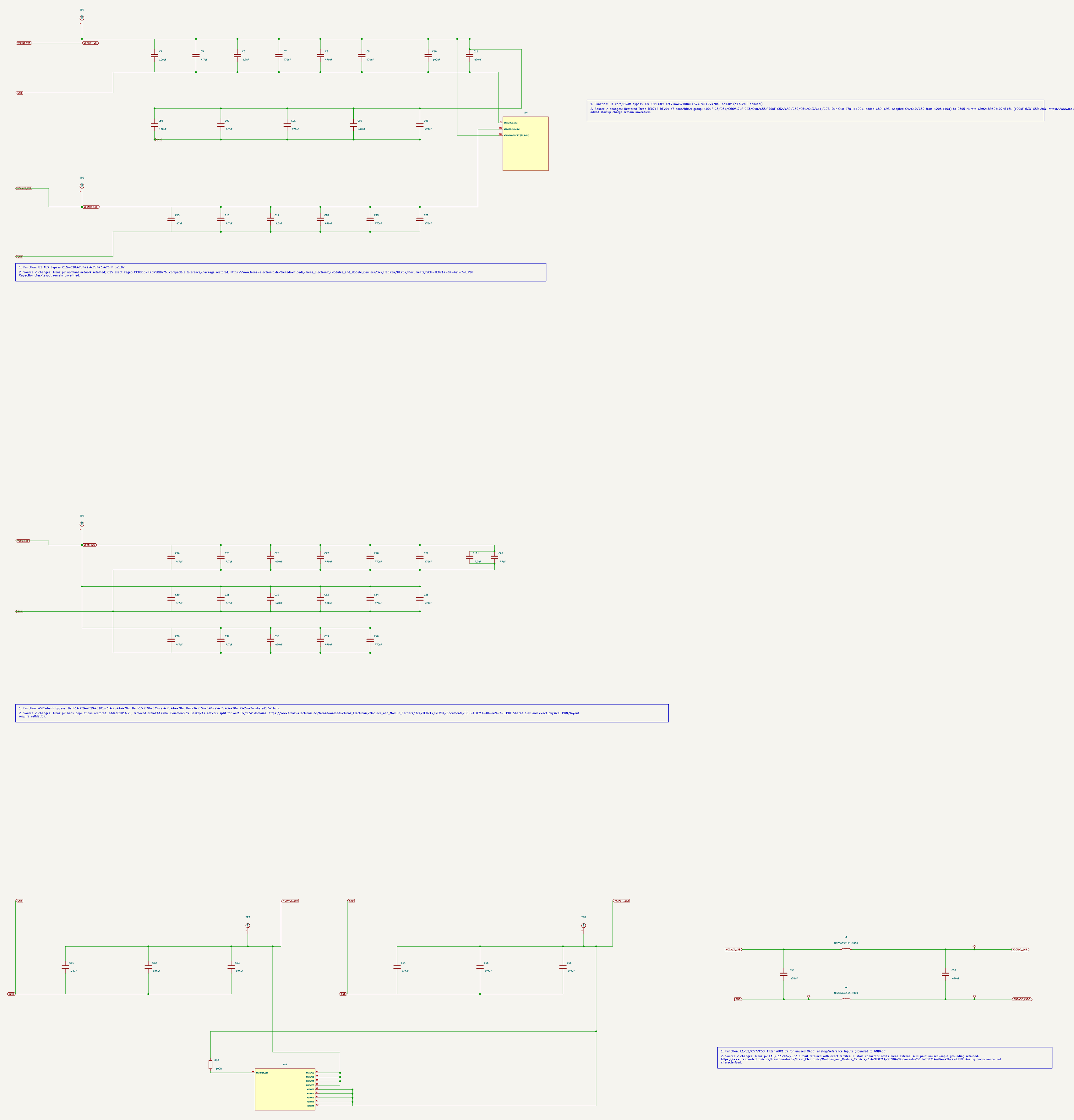


1. Function: U11 holds PROGRAM_B and INIT_B LOW through two independent open-drain outputs; R17/R18 release them to 1.8 V. U14 drives the 1.5 V active-low U8 enable. C82/C63 are local bypass.
2. Source / changes: Correspondence: Trenz p7 D1/D2 reset-hold functions -> U11; U14 is added for our U8 flash translator, not Trenz boot-mode inverter U9. V64 electrical network unchanged. Codex voltage adaptation of Trenz p7 D1/D2 reset function; inherited V61 agreed flash/U8 support retained. Trenz diode/common3.3V solution is not used across our1.8V/1.5V domains. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF https://assets.nexperia.com/documents/data-sheet/74AUP2G07.pdf https://www.ti.com/lit/ds/symlink/sn74aup1g14.pdf

1. Function: U1 configuration: Master SPI001,CFGBVS GND, Bank0/JTAG1.8V; ASIC banks1.5V. PROGRAM_B and INIT_B reset independently. VCCBATT E11 now AUX1.8V.
2. Source / changes: Trenz p7 values restored where compatible: R17/R18/R19/R22=4.99k; E11 GND->AUX. Mode ties/1.8V rails and x1 flash remain agreed adaptations.
https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF AMD UG470v1.17 Table2-4 permits AUX for unused BBRAM.
https://docs.amd.com/api/khub/documents/FOs3lXmlcWxBhTlFxVKyGA/content

1. Function: Bank0 bypass C59,C94–C100:47uF+3x4.7uF+4x470nF at1.8V, separate from1.5V Bank14.
2. Source / changes: Codex split of Trenz p7 shared Bank0/14 network to preserve agreed voltage domains; addedC94–C100, C59 exact Trenz CC0805MKX5R5BB476.
https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF Duplication adds startup charge; effective capacitance and physical placement require validation.

05. FPGA SUPPLY PINS + BYPASS + XADC | Trenz p7



1. Function: U1 core/BRAM bypass: C4–C11,C89–C93 now3x100uF+3x4.7uF+7x470nF on1.0V (317.39uF nominal).
2. Source / changes: Restored Trenz TE0714 REV04 p7 core/BRAM group: 100uF C8/C54/C56;4.7uF C43/C48/C55;470nF C52/C49/C50/C51/C13/C11/C27. Our C10 47u->100u, added C89–C93. Adapted C4/C10/C89 from 1206 (10%) to 0805 Murata GRM21BR60J107ME15L (100uF 6.3V X5R 20%, https://www.mouser.com/datasheet/2/281/1/GRM21BR60J107ME15_01A-1986857.pdf) for V6 size reduction. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF DC bias, layout, PDN and added startup charge remain unverified.

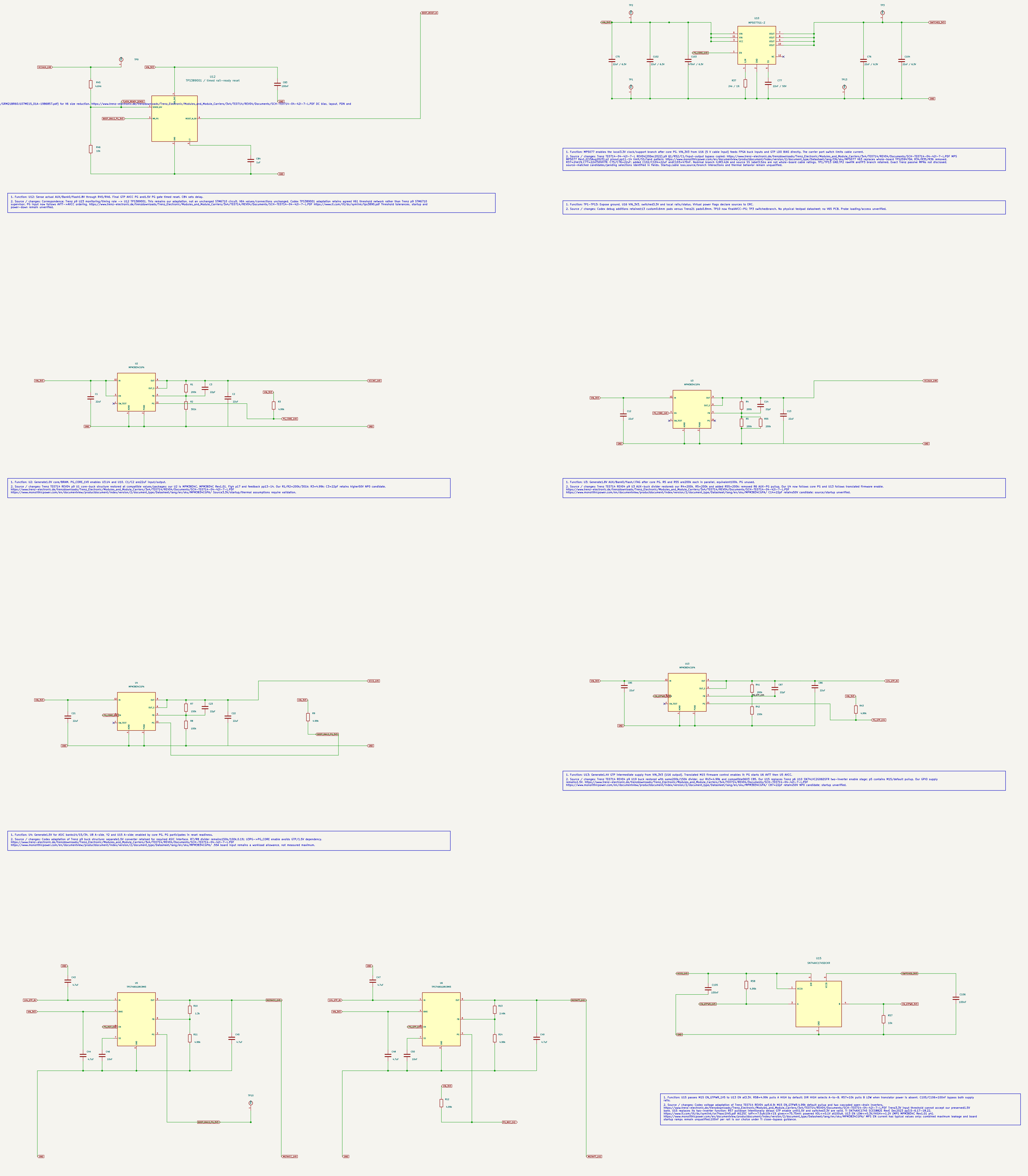
1. Function: U1 AUX bypass C15–C20:47uF+2x4.7uF+3x470nF on1.8V.
2. Source / changes: Trenz p7 nominal network retained; C15 exact Yageo CC0805MKX5R5BB476, compatible tolerance/package restored. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF Capacitor bias/layout remain unverified.

1. Function: ASIC-bank bypass: Bank14 C24–C29+C101=3x4.7u+4x470n; Bank15 C30–C35=2x4.7u+4x470n; Bank34 C36–C40=2x4.7u+3x470n. C42=47u shared1.5V bulk.
2. Source / changes: Trenz p7 bank populations restored; addedC1014.7u; removed extraC41470n. Common3.3V Bank0/14 network split for our1.8V/1.5V domains. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF Shared bulk and exact physical PDN/layout require validation.

1. Function: GTP bypass C51–C56: each analog rail now4.7uF+2x470nF. R16=100R connects MGTRREF to MGTAVTT.
2. Source / changes: Trenz p7 nominal/value/package/tolerance restored: C52/C53/C55/C56 100n->470n; C51/C54 4.7u20%; R16 selected100R0.1%25ppm ERA1A meets source100R1%TK100 class. Selected compatible Murata/Yageo candidates; Trenz exact MPNs not all disclosed. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF AMD UG482 v1.9 Table5-7 remains a design reference; actual PDN unverified.

1. Function: L1/L2/C57/C58: Filter AUX1.8V for unused XADC; analog/reference inputs grounded to GNDADC.
2. Source / changes: Trenz p7 L10/L11/C62/C63 circuit retained with exact ferrites. Custom connector omits Trenz external ADC pair; unused-input grounding retained. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF Analog performance not characterized.

06. REGULATORS + READINESS SUPERVISOR | Trenz p8 notes



1. Function: U12: Sense actual AUX/Bank0/flash1.8V through R45/R46. Final GTP AVCC PG and1.5V PG gate timed reset. C84 sets delay.
2. Source / changes: Correspondence: Trenz p9 U23 monitoring/timing role -> U12 TPS389001. This remains our adaptation, not an unchanged STM6710 circuit. V64 values/connections unchanged. Codex TPS389001 adaptation retains agreed V61 threshold network rather than Trenz p9 STM6710 supervisor. PG input now follows AVTT->AVCC ordering. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF https://www.ti.com/lit/ds/symlink/tps3890.pdf Threshold tolerances, startup and power-down remain unverified.

1. Function: MP5077 enables the local3.3V clock/support branch after core PG. VIN_3V3 from U16 (5 V cable input) feeds FPGA buck inputs and GTP LDO BIAS directly. The carrier port switch limits cable current.
2. Source / changes: Trenz TE0714-04-42I-7-L REV04(20Dec2022),p9 Q1/R52/C1/input-output bypass copied: https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF MPS MP5077 Rev1.2(15Aug2023),p2 pinout,pp11-14 limit/SS/land pattern: https://www.monolithicpower.com/en/documentview/productdocument/index/version/2/document_type/Datasheet/lang/EN/sku/MP5077 V63 replaces whole-board TPS259470A; R34/R35/R36 removed. R37=24k1%,C77=22nF50VX7R; C75/C76=22uF; added C102/C104=22uF andC103=470nF. Nominal branch ILIM3.42A and source SS label3.5ms are not whole-board cable ratings. TP1/TP13 GND,TP2 rawVIN andTP3 branch retained. Exact Trenz passive MPNs not disclosed; source-matched candidates/pending selections identified in fields. Startup,cable loss,source/branch interactions and thermal behavior remain unqualified.

1. Function: TP1–TP13: Expose ground, U16 VIN_3V3, switched3.3V and local rails/status. Virtual power flags declare sources to ERC.
2. Source / changes: Codex debug additions retained;13 custom0.6mm pads versus Trenz21 pads0.8mm. TP10 now finalAVCC-PG; TP3 switchedbranch. No physical testpad datasheet; no V65 PCB. Probe loading/access unverified.

1. Function: U2: Generate1.0V core/BRAM. PG_CORE_1V0 enables U3,U4 and U10. C1/C2 are22uF input/output.
2. Source / changes: Trenz TE0714 REV04 p9 U1 core-buck structure restored at compatible values/packages; our U2 is MPM3834C. MPM3834C Rev1.01, Fig4 p17 and feedback pp13-14. Our R1/R2=200k/301k; R3=4.99k; C3=22pF retains higher50V NP0 candidate.
https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF
https://www.monolithicpower.com/en/documentview/productdocument/index/version/2/document_type/Datasheet/lang/en/sku/MPM3834CGPA/ Source3.3V/startup/thermal assumptions require validation.

1. Function: U3: Generate1.8V AUX/Bank0/flash/JTAG after core PG. R5 and R55 are200k each in parallel, equivalent100k. PG unused.
2. Source / changes: Trenz TE0714 REV04 p9 U3 AUX-buck divider restored: our R4=200k, R5=200k and added R55=200k; removed R6 AUX-PG pullup. Our U4 now follows core PG and U13 follows translated firmware enable.
https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF
https://www.monolithicpower.com/en/documentview/productdocument/index/version/2/document_type/Datasheet/lang/en/sku/MPM3834CGPA/ C14=22pF retains50V NP0 candidate; source/startup unverified.

1. Function: U13: Generate1.4V GTP intermediate supply from VIN_3V3 (U16 output). Translated M15 firmware control enables it; PG starts U6 AVTT then U5 AVCC.
2. Source / changes: Trenz TE0714 REV04 p9 U19 buck restored with same200k/150k divider, our R43=4.99k and compatible0603 C85. Our U15 replaces Trenz p6 U10 SN74LVC2G06DSFR two-inverter enable stage; p5 contains M15/default pullup. Our GPIO supply remains1.5V. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF
https://www.monolithicpower.com/en/documentview/productdocument/index/version/2/document_type/Datasheet/lang/en/sku/MPM3834CGPA/ C87=22pF retains50V NP0 candidate; startup unverified.

1. Function: U4: Generate1.5V for ASIC banks14/15/34, U8 A-side, Y2 and U15 A-side; enabled by core PG. PG participates in reset readiness.
2. Source / changes: Codex adaptation of Trenz p9 buck structure: separate1.5V converter retained for required ASIC interface. R7/R8 divider remains150k/100k,0.1%; U3PG->PG_CORE enable avoids GTP/1.5V dependency.
https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF
https://www.monolithicpower.com/en/documentview/productdocument/index/version/2/document_type/Datasheet/lang/en/sku/MPM3834CGPA/ .55A board input remains a workload allowance, not measured maximum.

Sheet: /
File: Artix_Frontend_V29.kicad_sch
Title: FPGA V54 dual60 direct FPGA/LDO interface – unrouted
Size: User
Date: 2026-10-19
Rev: V54
KiCad E.D.A. 10.0.6
Id: 1/1